Environmental Innovation and Societal Transitions 50 (2024) 100808

Contents lists available at ScienceDirect

Environmental Innovation and Societal Transitions

journal homepage: www.elsevier.com/locate/eist

Research article

Improving multilevel policy mixes for sustainable urban mobility transition

Xu Liu [a,*], Marc Dijk [a], Carlo Colombo [b,c]

[a] Maastricht Sustainability Institute, Maastricht University, Maastricht, the Netherlands
[b] Faculty of Law, Maastricht University, Maastricht, the Netherlands
[c] Centre for European Studies and Comparative Politics, Sciences Po, Paris, France

ARTICLE INFO

Keywords:
Policy mixes
Urban mobility transition
Sustainability transition
Multilevel governance

ABSTRACT

Transitioning to sustainable urban mobility in the EU requires a complex policy mix covering various transport modalities and levels of governance. Yet, existing studies focus mainly on a single modality or governance level as unit of analysis. To fill this gap, this article combines the literatures on policy mixes and on urban mobility transition to investigate the relations between various policies influencing the transition to sustainable urban mobility across governance levels. Analysing data from 32 interviews focused on Dutch cities, two key conflicts are identified as hampering the transition to sustainable urban mobility: i) a bias of national funding towards solving bottlenecks through infrastructure development; ii) a significant influence of the national level on the local level, while the former is hardly influenced by others. The analysis suggests that these conflicts might be overcome by shifting the emphasis of national funding to mobility in a broader sense and institutionalizing multilevel cooperation.

1. Introduction

To achieve the goal of climate neutrality by 2050, the EU recently raised its level of ambition in the transport sector, setting a target of a 90 % reduction in CO2 emissions (EU, 2019). Modelling of abatement options, however, indicates that new, cleaner propulsion technologies (e.g., electric vehicles) will not suffice to reduce current transport emissions to the required level (EC, 2021). Rather, car trips need to be replaced by public transportation, and for shorter distances, investment and promotion of 'active travel', such as walking, cycling and e-biking, is necessary (Brand et al., 2021).[1] Given the tremendous scale of this challenge, European cities, in particular in the Netherlands, are raising their level of policy ambition to promote sustainable urban mobility transition (te Boveldt et al., 2022; GA, 2019; GDH, 2022). We define sustainable urban mobility transition as a reconfiguration process of the urban mobility system (Laakso et al., 2021) which results in a significant reduction in the share of car mobility (Banister, 2008; Cairns et al., 2014) while ensuring a high and socially equal level of accessibility.

Lately, various approaches and policies have been implemented to promote a transition to sustainable urban mobility by policy-makers from the urban to the regional, national and European levels. Over time, this has led to a 'mix of policies' (Howlett and Rayner, 2007) affecting urban transport, including policies on car transport, public transport and non-motorized transport. Their effects, however, have been limited and scattered. Although private car use has been successfully reduced in a few large European cities, in

* Corresponding author.
E-mail address: xu.liu@maastrichtuniversity.nl (X. Liu).

[1] Avoiding the need to travel in the first place (for instance through the 15-minute-city concept) is another possible strategy.

https://doi.org/10.1016/j.eist.2023.100808
Received 18 February 2023; Received in revised form 8 December 2023; Accepted 20 December 2023
Available online 30 December 2023
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many others, it has grown (ECA, 2020). At the same time, sustainable mobility planning capacities have remained weak in smaller cities and towns (Dragutescu et al., 2020).

Research on sustainable transitions (Köhler et al., 2019; Kivimaa and Kern, 2016; Reichardt et al., 2016; Schmidt and Sewerin, 2019; Zepa and Hoffmann, 2023) suggests that attention to deliberate policy mixes, i.e. bundles of individual policy instruments or tools and techniques used by government in order to achieve their policy goals (Howlett, 2005), is critical to understand the degree to which policies facilitate (or hinder) transitions. Yet, most literature on urban transport policy tools has so far considered the effects of less complex designs, often concerning one instrument, transport modality and one governance level.[2] At the same time, studies on policy mixes and their effects on sustainable transition have focused on national and supranational energy issues, leaving the overall (in)coherence and (in)effectiveness of the urban mobility mix of policies underexposed (Howlett et al., 2017; Dijk et al., 2018).[3]

Building on del Río González (2007) and Givoni et al. (2013), our research assumed three possible relations between policies:[4] (1) synergetic linkages, in which policies reinforce or complement one another in achieving changes towards sustainable mobility; (2) contradictory or conflicting linkages, in which two or more countervailing policy measures undermine the functional capacity of both or either to support sustainable mobility; and (3) 'neutral' relationships, in which neither of the above applies or the effect is mixed or unclear. A policy mix is deemed to be more coherent when it has more synergetic linkages and less contradictory linkages between policies.

The aim of this article is to develop a sharper understanding of the main conflicts and synergies between policies concerning sustainable urban mobility across the various governance levels. It then makes a start in exploring how these conflicts may be overcome and synergies be promoted. Specifically, it addresses two main questions: (1) What are the key conflicts between policies for sustainable urban mobility transitions in a multilevel governance context? (2) How can these conflicts be overcome in order to improve policy outputs and outcomes? Empirically, our study is based on 32 interviews with policymakers at four governance levels, focusing on the case of cities in the Netherlands. In short, we sought to learn what they saw as the most relevant instruments for sustainable urban mobility transitions, alongside the effects of these instruments and relations.

In answering these questions, our study contributes to both academic and professional knowledge on urban sustainability transitions. First, it advances academic research on both sustainable transitions and urban mobility policy by pioneering a novel connection between two debates – that on (multilevel) policy mixes and that on sustainable urban transitions – as urged by previous authors on this topic (Hodson and Marvin, 2010). Second, the research offers practical insights and recommendations for professionals to improve the vertical policy mix for sustainable urban mobility in Europe.

The article is structured as follows. Section 2 develops the conceptual frame for the analysis, building on previous research. Section 3 describes the methodology. Section 4 presents the findings, while Section 5 discusses implications for policy practitioners and research. Finally, Section 6 presents conclusions.

2. Towards a multilevel understanding of policy mixes for sustainable urban mobility transitions

2.1. Urban transition

Our definition of sustainable urban mobility, presented above, is in line with the more general framing of urban transition by (Hodson et al., 2017). Accordingly, urban transitions are not about technological or social innovations or their impacts per se, but about how multiple innovations, including those in policy and planning, shape patterns or forms of reconfiguration (entailing 'the new' and 'the established'). Attention to urban transitions has grown in the past decade, particularly in the context of debates on socio-technical transitions and urban geography. Key themes in these debates are the spatial context, scales and urban experiments. First, although early transition studies (roughly before 2012) focused largely on the national level, neglecting the spatial context, a number of later transition studies has sought to better incorporate spatial aspects (Coenen et al., 2012; Hansen and Coenen 2015). In this regard, some authors have focused on local energy issues (Späth and Rohracher, 2012), while others have considered local climate initiatives more generally (Bulkeley and Castán Broto, 2013).

Second, regarding scale, scholars have stressed that urban transitions are shaped by developments at multiple scales beyond the local (Coenen et al., 2012). This implies that, to analyse urban transitions, it is not enough to account for the distribution of activities within a particular space, such as an urban area. Instead, according to this scholarship, there is also a need to examine the interconnections with other places (Bridge et al., 2013) as well as with territorially unspecific factors such as national and international

[2] A few studies analyze (or offer concepts for analysis) of more complex transport policy mixes. Yet, they do so only for policies adopted at the urban level: Curtis and Low 2012; Givoni et al 2013; Givoni 2014; Buehler et al. 2017; Dijk et al 2018.

[3] Zepa & Hoffmann's (2023) study is an exception in this regard. It analyses how sustainable energy policy mixes in Latvia develop across vertical scales of governance (i.e., local, national, European). To this end, the study considers the consistency, coherence, comprehensiveness, and credibility of vertical policy mixes. The findings highlight how frequently changing strategies and policy instruments at the state and local levels in Latvia offset the ambitious sustainability agenda at the European level, reinforcing the status quo against substantial political and economic changes.

[4] In addition to synergetic and contradictory relations, Givoni et al (2013) distinguish pre-condition relations. These involve situations whereby the successful implementation of one policy measure remains wholly contingent upon the prior successful implementation of another. We consider this as a special case of synergistic linkages, as the latter includes all cases in which the functional capacity of one policy measure is enhanced by the presence of another measure. Consequently, for our purpose, identified cases of pre-condition relations will be classified under the more general category of synergistic linkages.

institutions and networks of interests (Hodson et al., 2013).

Third, an expanding range of studies has looked at urban experimentation in relation to urban transitions, drawing various observations (Evans and Karvonen, 2014). Some have stressed the ability, or potential, of experiments to create highly context-sensitive and locally-relevant knowledge (Karvonen and Van Heur, 2014). The definition of experiments, in this regard, has been rather loose, and analyses of impacts have tended to involve a rather unspecific notion of 'change' (ibid.). The study of urban experiments has spanned various fields, with important contributions found in the sustainability transitions literature (see, e.g., Torrens et al. 2019, Huang and Broto, 2018, Frantzeskaki et al., 2017) and the urban geography and governance literature (see, e.g., Bulkeley et al., 2011, Scholl and De Kraker, 2021). Hodson et al. (2017) observed that to contribute to urban transitions, the focus of experiments must go beyond the experimental practices and also involve pre-existing infrastructures and institutions, including those at multiple levels of governance.[5]

Yet, insofar as empirical studies of urban transition are concerned, little research has taken a 'whole system reconfiguration' approach (McMeekin et al., 2019). Instead, most have focused on a single urban experiment concerning some radical technology or social innovation with a narrow group of practitioners from a city-wide perspective. Exceptions are historical transition cases, such as Switzer et al. (2015) and Dijk et al. (2021), with the latter also highlighting the role of policies in sustainability transitions. Nonetheless, the ways in which mixes of policy goals, instruments and processes interact seems of crucial importance in the degree to which policy facilitates (or hinders) urban transitions (Switzer et al., 2015; Dijk et al., 2021). We zoom in on this now.

2.2. Policy for sustainable mobility

2.2.1. Policies across multiple governance levels

As noted in Section 1, the EU system has multiple governance levels, with the European, national and subnational policy arenas embedded in one another (Kern and Bulkeley, 2009; Bache and Flinders, 2004). The notion of multilevel governance can be conceived in a narrow and a broader sense (Kern and Bulkeley, 2009). In the narrow sense, it refers to the distributed regulatory competences between local, national and supranational governmental institutions. In the broader sense, it also encompasses other sorts of formal and informal interactions between actors from different levels of governance, including public-private partnerships (Stephenson, 2013; Beisheim et al., 2010).

This article employs the broader definition. It does so because, in the past decades, public authority in Europe has not simply shifted upwards to European institutions, but also downwards, to multiple territorial levels and a variety of private and public actors (Kern and Bulkeley, 2009; Rosamond, 2007). This includes the transfer of public authority from national to subnational levels (i.e., regions, provinces and municipalities), due to processes of decentralization. The result is a rather complex situation of somewhat indistinct boundaries between the different governance spheres, with some policy actors (e.g., NGOs and lobby or branch organizations) being active across various levels at the same time. The different governance levels are not only affected by immediately adjacent levels (e.g., the EU and national levels), but there are also direct relations between EU institutions and local authorities (Marshall, 2005). For instance, Urban Innovative Actions is an initiative of the European Union that provides urban areas throughout Europe with resources and expertise to test new and unproven solutions to address urban challenges. Accordingly, there is a mix of policies across levels that aim to reconfigure urban transport in a more or less sustainable direction, and these policies may interfere with one another (Fig. 1).

In this rather complex policymaking context, Bemelmans-Videc and Vedung (1998) distinguished three forms of policy mixes[6] : 'vertical mixes', representing the implementation of policy measures aimed at policies at other levels of governance to achieve policy aims; 'chronological mixes', referring to implementation of policy measures in deliberate sequence, thus following a particular order in time; and 'horizontal mixes', meaning the implementation of two measures are aligned to achieve the same target and level of government. To keep the analysis manageable, our analysis will focus on vertical mixes. The question is how the coherence of policy mixes in this rather complex, multilevel governance context can be enhanced to promote sustainable urban mobility transition.

2.2.2. Mobility policy mixes

The transport policy literature suggests two options for reducing transport emissions: a shift to cleaner technology and behavioural change (Chapman, 2007). In practice, technological solutions have dominated policies on transport and climate change (ibid.). Research indicates, however, that at least some level of behavioural change is also needed if carbon emissions from transport are to be significantly reduced (Anable et al., 2012). For instance, a recent study shows that if policy could shift an average person to cycle 1 trip/day more and driving 1 trip/day less for 200 days a year, this would decrease mobility-related lifecycle CO_2 emissions by about 0.5 tonnes over a year (Brand et al., 2021). Accordingly, scholars have sought to widen the focus beyond one or a few instruments (e.g., information provision or reduced fares) and transcend the hard/soft and behaviour/technology dualisms (Schwanen et al., 2012). In this regard, the most relevant mobility policy studies are those that propose a 'modal shift', referring mainly to a (significant) reduction

[5] Ehnert et al (2018) and Servou et al (2022) are the only studies analysing urban experiments (or initiatives) in a multi-level governance context. The former explores the difference between federal and unitary policy systems in providing opportunities and obstacles for the transition to urban sustainability. The latter studies how multi-level governance cultures influence the co-creation of experiments and their impact. Both analyses, however, do not address (multi-level) policy mixes.

[6] In fact, in their study Bemelmans & Verdung use the term 'package' and not 'mix'. However, for the purposes of this study, we chose to use the term 'mix'. This preference aligns with the prevalent definition by Howlett, as mentioned earlier. In Howlett and Rayner's study, the term 'packaging' generally refers to the creation of newly designed mixes, as distinct from the restructuring of existing mixes (Howlett and Rayner 2013).

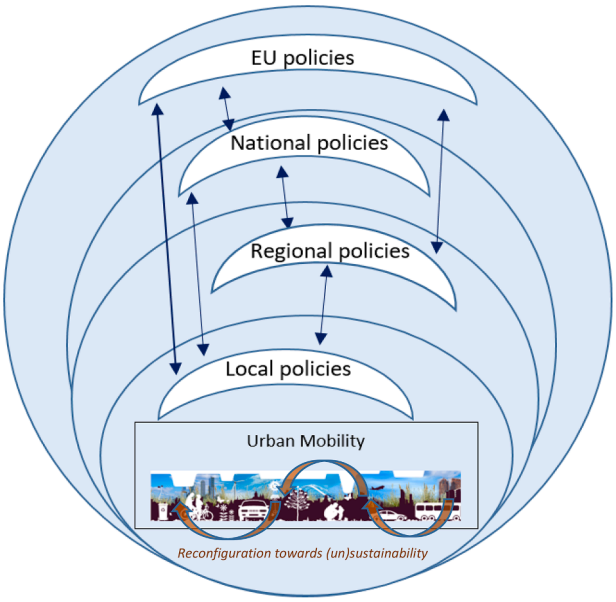

Fig. 1. Reconfiguration of urban mobility amid multilevel policy interaction (Source: Authors).

of the share of private car mobility. To achieve this, policies have traditionally employed 'push' and 'pull' measures (Batty et al., 2015; Curtis, 2018). 'Pull' strategies aim to increase the supply of high-quality, low carbon transport modes, such as public transport and active mobility (walking and cycling), and to expand infrastructure supportive of these modes (Buehler et al., 2017; Strompen et al., 2017). By contrast, 'push' strategies seek to discourage unsustainable behaviours (i.e., private car use) by making these forms of transport less convenient, for example, by implementing road-use tariffs, congestion charges and traffic slowing measures (Buehler et al., 2017; Strompen et al., 2017; Curtis, 2018). To most effectively encourage a modal shift, it is generally accepted that both 'push' and 'pull' measures are needed (IEA, 2009; Black and Schreffler, 2010; Strompen et al., 2017). Nonetheless, most transport literature addresses these different types of policies in isolation, considering, for example, either strategies to encourage active mobility (i.e., walking and cycling) or those to encourage public transportation or to constrain car use.

Transition scholars have also argued that to initiate a sustainability transition, policy mixes should combine instruments aiming both to create 'the new' and to destabilize or constrain 'the old' (Kemp et al., 1998; Kivimaa and Kern, 2016). This corresponds with the transport policy literature's attribution of the most successful modal shifts to combinations of 'push' and 'pull' measures. An example of such a strategy is reported by Buehler et al. (2017). They found that, in all five European cities under assessment, a modal shift was successfully encouraged through "a coordinated package of mutually reinforcing transport and land-use policies that, in combination, … made car use slower, less convenient, and more costly, while increasing the safety, convenience, and feasibility of walking, cycling, and public transport". Despite these successes, Curtis and Low (2012) observed that in many cities policymakers continued to struggle with conflicting imperatives to constrain and enable car use). Although some have developed visions of reduced shares of car mobility, they have as yet been unable to fully drop the car-enabling logic of 'predict-and-provide'; that is, predicting where (vehicular) congestion may arise in the future and adding infrastructural capacity (roads) to mitigate it. A study on urban mobility in Stuttgart similarly found contradictions between the measures needed for a sustainability transition and those that were feasible for policy-makers (Schippl and Arnold, 2020).

Due to the limited number of studies that combine push and pull measures, and the fact that most of those available address the local governance level only, the overall (in)coherence and (in)effectiveness of the multilevel urban mobility mix of policies remains underexposed (Howlett et al., 2017; Dijk et al., 2018). This article, therefore, analyses urban mobility transitions using a conceptual frame that connects synergies and conflicts in multilevel mix of policies with shifts towards sustainable mobility modalities (e.g., public transport, cycling and walking). The objective is to sharpen understanding of the main conflicts across policies to support sustainable urban mobility transitions at the various levels of governance and to elicit initial lessons on how these conflicts might be overcome.

3. Method and data

We chose a qualitative research method, interviewing, to answer the research question, because this method is more sensitive to the subtleties of policy effects and interactions between different governance levels than quantitative methods (Kivimaa et al., 2017). Specifically, we conducted semi-structured interviews with representatives of different governance levels and departments. Our aim was to tap into the particular knowledge, experiences and perspectives of each concerning policy impacts. Jointly, they provided a balanced view of the most influential factors and (in)coherencies of the mix of policies concerning sustainable urban mobility

development.

3.1. Interviews

We started by contacting representatives of the governance levels that had issued the most explicit urban mobility policy plans or visions. These were the city and European levels. At each of these levels, we contacted one governmental and one non-governmental organization: the Directorate-General for Mobility and Transport (DG MOVE) and the Cities and Regions for Transport Innovation (POLIS) Network at the European level and the Municipality of Maastricht and Maastricht Bereikbaar at the urban level. Combining suggestions from these interviewees ('snowballing') with our own analyses of stakeholders in associated research (positional approach), we selected further interviewees (Myers and Newman, 2007). We requested interviews in cities of various sizes and locations, with the final sample including Rotterdam (population about 590,000, located in the western Netherlands), Nijmegen (population about 185,000, located in the eastern Netherlands) and Maastricht (population about 125,000, located in the southern Netherlands). These cities, respectively, ranked 2, 10 and 23 among the country's 32 largest cities in terms of inhabitants. We then interviewed subjects at the regional and national levels.

We stopped contacting additional interviewees at each level when we reached data saturation, meaning that a subsequent interview delivered little or no new insights. This process yielded 32 semi-structured interviews with policymakers, programme managers, international organizations, and non-governmental organizations at the EU level (10), the national level (9), the regional level (6) and the local level (7). All interviews were conducted in 2020 and 2021 (Table 1, Appendix A provides further details on interviewees).

Although the three cities and regions in the Netherlands cannot be considered fully representative of Dutch cities overall, they do reflect different sizes of cities and, with hindsight, also different characteristics of city centres and street patterns. Rotterdam is modern, Maastricht is historic, and Nijmegen has a mixed character. The semi-structured interviews lasted between 40 and 100 min and were transcribed. Appendix B presents the interview guide. We began each interview by asking the respondent how 'sustainable urban mobility' was defined within their own department or organization. Subsequently, we asked what policies or instruments both at their level and at other governance levels they saw as having the most significant impact on urban mobility. We also asked each interviewee how they saw relationships between policies at the various governance levels, prompting them to think in terms of conflictual, synergistic and neutral relationships.

To better understand the context of the responses, we also accessed policy reports concerning the policies mentioned by interviewees (see list in Appendix C). The respondents explicitly mentioned some reports, while we identified others ourselves.

Table 1
List of the interviewees.

Level	No.	Organization
EU (10)	1	Local Governments for Sustainability (ICLEI)
	2	European Cyclists' Federation
	3	Directorate-General for Environment (DG ENV)
	4	Eurocities
	5	Directorate-General for Mobility and Transport (DG MOVE)
	6	DG MOVE
	7	Directorate-General for Regional and Urban Policy (DG REGIO)
	8	DG ENV
	9	Cities and Regions for Transport Innovation (POLIS)
	10	European Institute of Innovation and Technology (EIT) Urban Mobility
National (9)	11	Friends of the Earth Netherlands (Milieudefensie)
	12	Nature & Environment (Natuur & Milleu)
	13	Urban Mobility at German Institute of Urban Affairs
	14	Netherlands Environment Assessment Agency (Planbureau voor de Leefomgeving)
	15	Ministry of Infrastructure and Water Management
	16	Ministry of Infrastructure and Water Management
	17a&b	a. Ministry of Internal Affairs (Ministerie van Binnenlandse Zaken en Koninkrijksrelaties) b. Ministry of Internal Affairs
	18	Ministry of Internal Affairs
Regional (6)	19	Rotterdam–The Hague Metropolitan Area
	20	Province Zuid Holland
	21	Province Limburg
	22	Province Gelderland
	23	Arnhem Nijmegen Region
	24	Arnhem Nijmegen Region
Local (7)	25	Maastricht Bereikbaar
	26	Rotterdam City
	27	Rotterdam City
	28	Rotterdam City
	29	Maastricht City
	30	Maastricht City
	31	Nijmegen City

3.2. Interview analysis

All interviews were recorded and transcribed. Then the software tool Atlas.ti was used to systematically and efficiently code the interviews (Konopásek, 2007). The coding process began with each of the three authors coding two interviews, taking as potential codes the topics in the interview guide (i.e., 'definitions', 'policies', 'relations', 'patterns', 'actors', 'challenges' and 'solutions'). Then the authors met to compare and align the coding. In the end, three main coding categories were agreed on: 'definition of sustainable urban mobility', 'policies' and 'relations between different levels'.

In the next step, each author coded four interviews, one from each governance level. This was followed by another meeting to present and explain the codes defined. This again helped to align the coding process, although it did not lead to revision of the three main categories. Several sub-categories were agreed. Concerning the coding category 'definition of sustainable urban mobility', two sub-codes were applied: 'broader definitions' and 'narrower definitions'. This reflects the fact that there is no universal definition of sustainable urban mobility, either among experts and academics or among practitioners (Gudmundsson and Regmi, 2017; Marsden et al., 2010). However, these two general groups can be distinguished in the range of definitions in the academic literature (Kayal et al., 2014). The broader definition regards sustainable urban mobility as an alternative to conventional transport planning. The latter seeks mainly to accommodate travel demand at low cost, while the former aims to balance travel demand (i.e., 'accessibility') with quality of the urban space and equity for residents (i.e., 'liveability') and entails prioritization of active and public mobility over car mobility (Banister, 2008). The narrow definition refers to the 'greening' of transport, focused on reducing emissions, especially through low- or zero-emission vehicles (i.e., electrification or hydrogen vehicles) as well as low-emission zones. For the category 'policies', four sub-codes were applied reflecting the four governance levels discussed in Section 2: 'European policies', 'Dutch national policies', 'Dutch regional policies' and 'Dutch local policies'. By counting the number of times interviewees mentioned a policy, its degree of influence on urban mobility was determined. For the category 'relations between different levels', two sub-categories were applied, according to the types of relationships discussed in Section 1: 'synergetic' and 'conflictual'. By counting the number of interviewees that flagged a particular conflict or synergy, these were scored on degree of significance. In this fashion, all 32 interview transcripts were coded.

4. Results

This section presents the main results regarding definitions of sustainable urban mobility, policies affecting sustainable urban mobility, and lastly, the reported conflicts and synergies between the different policies adopted at various levels.

4.1. The definition of sustainable urban mobility

Concerning the definition of sustainable mobility, we analysed similarities and differences between and also within the four governance levels (supranational, national, regional and local), highlighting consistencies and inconsistencies in definitions and goals of sustainable urban mobility.

4.1.1. Consistencies

At the Dutch national, regional and local levels, interviewees were highly consistent in defining sustainable urban mobility in the broader sense, that is, as a modal shift that implies less car mobility. They mentioned that CO_2 reduction is increasingly cities' biggest concern, triggered by both the European Green Deal and the Dutch National Climate Agreement (NCA, 2019). In addition, city liveability is gaining in prominence, including safety and health aspects, driving regional and local governments to work together to promote sustainable urban mobility.

Interviewees from two European NGOs stressed that climate change is a strong motivator for promoting sustainable urban mobility, which they broadly defined as facilitating a modal shift to public transport, walking and cycling, with an eye for safety, accessibility and affordability as well (interviewee 9 and 10). These interviewees also noted that urban mobility should be integrated into all relevant policy domains (e.g. energy policies, environment policies, data policies, privacy policies, etc.), rather than being limited only to the field of transport and traffic regulation. The POLIS representative observed that many higher level (supranational and national) policymakers tended to neglect the complexity at the local level and noted trade-offs between different measures and policies. Moreover, this interviewee mentioned the need for a better approach to evaluate the impacts of a mix of policies (interviewee 9).

4.1.2. Inconsistencies

At the European level, some differences were found between departments and organizations. An interviewee working on urban environmental policy at the EU directorate-general for environment (DG ENV, interviewee 8) comprehended sustainable urban mobility narrowly as reducing air pollution by replacing fossil fuel-driven vehicles with clean vehicles. This interviewee commented that most current cars were designed for long distances and high speeds, but the type of vehicle best suited for travel within cities is something between a car and an electric bicycle, for which there is currently no legislative space. An interviewee from the EU Directorate-General for Mobility and Transport (DG MOVE, interviewee 5) said that sustainable urban mobility is about how to find usable, affordable, sustainable and alternative vehicles to replace traditional fossil-fuel cars. Yet, the other participant from the same department (interviewee 6) defined sustainable urban mobility more broadly, as urban transitions towards mobility with less impact on nature while meeting the needs of people and increasing quality of life.

4.2. Policies affecting sustainable urban mobility

When interviewees were asked which policies mainly affected sustainable urban mobility, they provided a list of 72 policies in total, ranging from programmes to instruments and measures (Table 2). To structure these, we grouped them by governance layer and objectives. This yielded five additional categories: (1) policy directives and plans that give direction to lower-level governments or departments (Nordholm, 2016); (2) car mobility policy measures seeking to alter car mobility, such as the creation of zero-emission zones or low-speed zones, the construction of new roads and tunnels, and the development of parking areas (Dijk et al., 2018); (3) public transport policies, such as extension of bus or tram lines, introduction of zero-emission buses and creation of fast bus lines; (4) non-motorized transport policies, consisting mainly of policy measures affecting walking and cycling (and e-steps); and (5) informative policies, meaning laws and regulations concerning information creation, processing, flows, access and use Braman (2011) such as the creation of expert groups to support policymaking.

The policies mentioned by the interviewees did not all fit perfectly into one group, though this was not critical for our analysis. For instance, some policies primarily affected car mobility, but also had some impact on other modalities. Table 2 presents the primary

Table 2
Policies related to sustainable urban mobility development mentioned by interviewees.

Types of policy	EU (24)	National (16)	Regional (13)	Local (19)
Mobility Development Directive & Plan	European Green Deal^{+}(11)[a] European Climate Pact^{+}(4) Sustainable Urban Mobility Plan (SUMP)$^{+}$(11) Clean and Better Transport in Cities^{+}(CIVITAS) (4) EU Air Quality Directive^{+}(10) EU White Paper on Transport^{+}(1) Urban Mobility Package^{+}(5) Urban Agenda^{+}(5) Smart and Sustainable Transport Strategy^{+}(1) Regional Development Funds$^{\pm}$(1)	Klimaatakkord^{+} (Climate Agreement) (14) Omgevingswet Agenda^{+}(1) Program Mobility as a Service^{+}(1) Multi-Year Programme for Infrastructure, Spatial Planning, and Transport (MIRT)$^{\pm}$(11) New Urbanization Strategy^{+}(1) National Strategy on Spatial Planning and the Environment (NOVI)$^{+}$ (2) Program MoVe^{+}(2) Smart Mobility^{+}(2)	POVI (Provincial Spatial Vision)$^{+}$(1) BDU$^{\pm}$ (Traffic and Transport Act) Funding (1) Smart Ways^{+}(1) Smart and Clean Mobility^{+}(1)	City Climate Agreement^{+}(1) Sustainable Procurement Policy^{+}(1) GOVI^{+} (Local Spatial Vision) (1) Stedelijk Verkeersplan^{+} (Urban Traffic Plan) (1)
Car Transport Policy	EU Car Emission Regulation^{+}(2) EU Emission Norms^{+}(1) EU Infrastructure Funding (TEN-T)$^{\pm}$(6) EU Alternative Fuel Infrastructure^{+}(1)	Logistical Laws^{+}(1) Car Taxation^{+}(5) National Truck Charging Plan^{+}(1) Charging Infrastructure^{+}(3)	Parking Norms^{+}(1) Car Sharing^{+}(1) Concession for Electric Vehicles^{+}(1) E-hubs^{+}(1) A50 Road Construction^{-}(1) A2 Tunnel Construction^{-}(1)	Zero-Emission Logistics Agreement^{+}(15) Low-Emission Zone^{+}(5) Speed-Limitation Zones^{+}(1) Congestion Charges^{+}(1) Parking Permit^{+}(1) Parking Taxation^{+}(1) P&R^{+} (Park and Ride) (1) E-hubs^{+}(1) Local Freight Hubs^{+}(1) EVs Charging Infrastructure^{+}(1) Narrowing Roads^{+}(2) Intelligent Traffic Lights^{+}(1)
Public Transport Policy	Public Service Obligations^{+}(1)	National Zero-Emission Bus^{+}(1)	Tendering Process for Public Transport Operator^{+} (7)	
Non-motorized Transport Policy	EU 100 Intelligent City Challenge^{+}(1) EU Driving Urban Transition Partnership^{+}(1) Green City Award^{+}(1) Mobility Week Campaign^{+}(1)	Beter Benutten Program^{+}(4)	South Limburg Bereikbaar^{+}(1) Fast Bike Lane^{+}(1)	Cycling Ambassador^{+}(1) Urban Labs^{+}(3) Investment in Cycling Infrastructure^{+}(3)
Informative Policy	EU Inter Service Consultation Groups^{+}(1) Open Data Directive^{+}(1) Shared Management Approach^{+}(1) Expert Group on Urban Mobility^{+}(1) European Innovation Bank^{+}(1)	Urban Logistics Expert^{+} (1) Pool Collaboration Project^{+}(1)		

Notes: ± To some extent supports/hinders a sustainable urban mobility transition (SUMT); + Supports SUMT; − Hinders SUMT. Numbers in parentheses indicate how many times the policy was mentioned by interviewees.

effect of the mentioned policies in supporting or hindering a sustainable urban mobility transition, according to the interviewees or reference documents. The table presents in parentheses the number of times a policy was mentioned by interviewees.

4.2.1. Main policies

The policies most frequently mentioned at the EU level were the European Green Deal, Sustainable Urban Mobility Plans (SUMPs), the Air Quality Directive and funding provided under the Trans-European Transport Network (TEN-T). Appendix D presents in brief how the interviewees described these policies, including their primary effect.[7] Table 2 further includes the main instruments mentioned by interviewees at the national level (the National Climate Agreement; the Multi-Year Programme for Infrastructure, Spatial Planning, and Transport (MIRT); and car taxation), the regional level (the tendering process for public transport operator) and the local level (low-emission zones and local traffic and land-use policies).

It is notable that the most explicit urban mobility policies at the EU level, specifically the SUMPs, the CIty-VITAlity-Sustainability (CIVITAS) programme and the Urban Agenda, are rather soft measures. Respondents described these as useful for developing knowledge and skills and inspiring other cities, but commented that participation remained voluntary (interviewee 1, 5, 6, 23). TEN-T funding was financially significant for large infrastructure projects, but its effect on sustainable mobility was considered slightly negative or indeterminate at best, due to its predilection towards investments in roads. One of the main policies at the national level, the National Climate Agreement, was developed to help meet the EU's Paris Agreement commitments. In similar vein, the Dutch national government has adopted new strategies and targets corresponding with the European Green Deal and European Climate Pact. A number of these were considered to have significant impact on regional and local mobility developments. 'Push' policy measures (e.g., taxation and emission regulations), mainly in the form of directives and car transport policies, were attributed stronger impacts than 'pull' measures (e.g., the zero-emission bus programme). At the regional and the local level, policy measures are specifically targeted at managing car use and public space. This is understandable, as these governance levels have direct authority over those particular areas. All regional policy measures were said to have mild influence. Hence, local governments had the most policies to limit car use, such as by regulating road space and parking areas, charging congestion fees and developing electric car sharing schemes (e-hubs). Furthermore, relatively small experimental and pilot projects were being done at the local level to provide lessons for larger-scale measures (interviewee 28).

4.3. Contradictions and synergies between levels

Interviewees made a number of general observations concerning the multilevel mix of policies. First, they commented that many people at various levels were involved in or 'doing' urban policy (interviewee 5). This brought substantial need for alignment and integration, to avert the risk of misalignments and conflicting policies. Second, the different levels were said to carry out slightly different roles and responsibilities. Furthermore, the policies at the various levels were being enacted rather independently. Although interviewees did not mention a long list of contradictions between policies at the various levels, comments were made on missed opportunities for alignments and synergies (interviewee 2). Overall, the interviewees agreed that urban mobility was primarily shaped and determined at the local level, through a process of co-creation with local stakeholders (interviewee 1 and 31) influenced by urban politics. EU level policies were said to have limited effect on urban mobility, since these mainly consisted of 'soft' instruments. However, to operate effectively, cities needed to attract funding, knowledge and skills; hence, they had to seek out partnerships beyond their administrative borders (interview 4). To do so effectively, cities required ample coordination with the national level (concerning funding) and the European level (concerning knowledge and skills).

Based on all 32 interview transcripts, a number of conflicting and synergetic relations were flagged between policies at various levels (Fig. 2). The main interactions were defined as those mentioned by at least three interviewees, of which four conflicts and nine synergies were identified. Appendix E reports all interactions mentioned by the interviewees.[8] We consider other implemented policies (that were not mentioned as conflicting or synergetic) as 'neutral' towards urban mobility transition.

4.4. Contradictions

The main conflict, flagged by seven policymakers at the national (1), regional (2) and local (4) levels, was that national funding did not really promote or was insufficient to promote urban sustainable mobility:

> *The local authority really doesn't have the funding to fix this kind of thing [*i.e.*, transport infrastructure and emission reductions]… so, you really have to look at the national and hopefully also the European level to help fund this. But it is really difficult. (interviewee 23, local level)*

[7] It is logical that instruments at EU and national level are mentioned more often, because all interviewees share national and EU level, but there are participants from three regions and cities with, at least partly, different instruments.

[8] Appendix E also shows that contested views on the nature of a number of relations (i.e. 8) emerged: some find it synergetic, others conflicting. Apparently, there is 'interpretative flexibility' in these cases. Instead, on the thirteen-relationships discussed in the main text no contestation emerged, except for one that is discussed below.

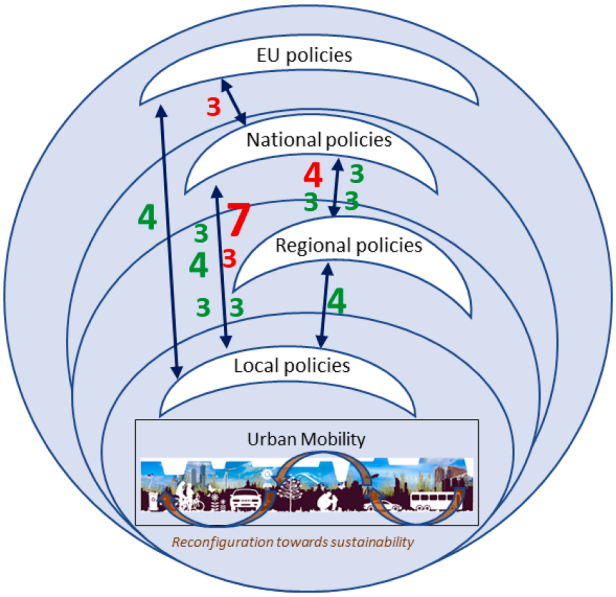

Fig. 2. The main contradictions and synergies reported.
(Source: Authors. Note: Red and green numbers represent the number of conflicts and synergies, respectively, between levels. For instance, the '**7**' between the national and local policies means that 7 interviewees flagged a conflict between these two levels.)

> *There's no monitoring [of implementation of the National Climate Agreement] from the national level. There's no extra money, except for some subjects, such as charging infrastructures. [There's] no extra money for reaching the [sustainable urban mobility] goals. (interviewee 28, local level)*

> *The policy framework of the ministry, for example, also excludes everything that's outside of their framework. So, even though some measures, some ideas, have a very positive contribution to, for example, sustainability. There won't be any funding available. (interviewee 25, regional level)*

Although the Dutch national government offers funding to the cities, in many cases, resources are provided only for projects that serve national priorities, which centre on 'solving congestion' and 'reducing vehicle emissions'. Moreover, in some cases, cities have to compete with each other to obtain the funding, as noted by a local mobility policymaker in Rotterdam:

> *When it comes to negotiating with the national government, and there is a lack of money and choices need to be made, then it's getting difficult, because of course, you get competition between Rotterdam, The Hague, etc.... There can be differences in priorities. (interviewee 26, local level)*

Three regional policymakers flagged a similar conflict at the regional level, observing that the funding which regions receive from the national government is actually mostly for road and housing-related infrastructure projects (interviewee 20, 21 and 22). For instance, an urban planner from South Holland (interviewee 20) pointed out:

> *We now get some funds from the Ministry of Internal Affairs. It's mainly for housing but not for mobility… The only link with mobility is that municipalities can also hire people to make sustainable mobility plans for new housing developments. (interviewee 20, regional level)*

An interviewee from the Province of Limburg commented on this issue as well:

> *We as a province try to look at mobility as a whole and not specifically on an infrastructural basis. And the ministry does take our suggestions into consideration. So, they are trying to change… But because now, the national government has this infrastructure fund, they still spend more on infrastructure than on broad mobility solutions. (interviewee 21, regional level)*

Another conflict flagged by European and national level interviewees is the EU's lack of authority or influence on national policies. An urban planner from the Netherlands Environmental Assessment Agency argued:

> *In the field of transport, European policies, except the car transport policies, have very little influence on [national] policies concerning cities. (interviewee 14, national level).*

This interviewee further noted that although regulations such as the EU's vehicle emission norms played a strong role, soft measures like the SUMPs had hardly any impact at the national governance level. Another example was given by the interviewee from Eurocities:

> *You can see in Central and Eastern Europe, there's still a big interest in using money from the European Commission to invest in highways…, which negatively affects mobility in the urban regions that these highways pass by, which traditionally have good public transport systems.… Member states still have a big say on how the EU money is spent. (interviewee 4, EU level)*

Thus, for some European countries, investing in road infrastructure instead of developing sustainable transport options still dominates national mobility policy, although this contradicts European level urban mobility strategies. European policymakers, moreover, have little room to change this.

Similarly, local level representatives complained that they lacked enough power to make big decisions, for which they were always dependent on the national government:

> *I also noticed that the big decisions are being made on a higher level. And in terms of financing, and the allocation of resources available, it is on the provincial and national level where the big decisions are being made. (interviewee 30, local level)*

A mobility innovator at DG MOVE observed the same problem in other member states as well:

> *The problem with going in a legal way to give more power to cities lies with member states… Our interlocutors are basically member states and the European Parliament but not cities. Therefore, every attempt by us to give more power to cities is normally met with strong opposition by member states, who basically want to control what is happening in their own territory and they don't want to share the power with cities. (interviewee 6, EU level)*

4.5. Synergies

At the same time, many interviewees saw synergies between the various governance layers. Three interrelations were especially prominent in this regard, each being mentioned by four interviewees. First, interviewees at both the European and the local level saw EU funding as well aligned with and supportive of local efforts. A team leader at DG REGIO elaborated the function of EU funding as follows:

> *The European budget, like the European Regional Development Fund,… can help people [cities] to get a little bit off their own block to connect with other cities to try something. (interviewee 7, EU level)*

The EU's sustainable mobility projects help cities develop their own skills and tools. An interviewee from Rotterdam mentioned the utility of these:

> *For example, with [our] big cities, we always check if we could have funding from the European Investment Bank, such as for the charging infrastructure. (interviewee 28, local level)*

Second, both national and local level representatives argued that national and local goals concerning urban mobility were increasingly in line (interviewee 12, 22, 29 and 31). For instance, the national Beter Benutten programme supported behavioural changes, such as alternatives to car trips, instead of (road) infrastructure development:

> *In 2012, the ministry implemented Beter Benutten. Before, the ministry every year spent billions of euros on infrastructure. But they came to the point that they said okay but maybe we shouldn't implement more infrastructure, but think about how to utilize the existing infrastructure better. (interviewee 25, local level)*

> *Beter Benutten… is more [about] efficiency.… And now it's transforming a little bit more into sustainable mobility. (interviewee 12, national level)*

Third, interviewees at the regional and local levels considered their goals to be well aligned. An example was provided by the interviewee from the Province of Limburg:

> *If we take a look at the cooperation with cities, for example, in Limburg, we have our global vision on mobility… Together with these cities, we make local policies… we come up with projects and policies to contribute to this vision and the goals we set in these visions to achieve within a couple of years. (interviewee 21, regional level)*

Similarly, an interviewee from Rotterdam noted alignment between local policies and the Province of South Holland's climate agreement:

> *The provinces have real targets from the climate agreement… and [it] is a new task of the provinces to organize vehicle charging in regions. Rotterdam [had the same goal and took the lead to] organize it for [other cities]. (interviewee 28, local level)*

Interviewees flagged other synergies as well. Both national and local representatives saw growing collaboration between the two levels (mentioned by three interviewees from both levels), hinting at joint urban development programmes. In particular, respondents mentioned the 'MoVe' programme, in which three governance levels (the Dutch national, regional and local levels) worked jointly on new spatial developments, combining responses to the housing shortage with solutions to mobility challenges ('accessibility'):

> *Despite the different roles and responsibilities at the various governance levels, we are trying to act like one government, one team with representatives of all three levels [national, regional and local]. (interviewee 17a, national level)*

> *[The national government] has a national strategy for charging infrastructure, and they work together with the bigger regions in the Netherlands… And they make their regional charging strategy. They are helping cities to make their own strategies and implement it. (interviewee 31, local level).*

Both national and regional interviewees had experienced growing collaboration between the two levels. This was noted by three interviewees from both levels:

> *We have an agreement with the Ministry of Infrastructure, that North Brabant pays 17 million. And we also pay 17 million for them. Then, all of us, national, regional and local, work together to make more use of public transport, to make more use of high-velocity bike lanes, … to make the role of sustainable mobility more clear. (interviewee 22, regional level)*

National and regional policymakers, too, mentioned shared goals:

> *We are working together more with the Ministry of Infrastructure. It is not only [that] we get money and then say goodbye. We find it very important that we also have our targets on sustainable mobility and we make deals together. (interviewee 22, regional level)*

Finally, although a lack of national financial support for promoting sustainable mobility transitions was flagged as the main conflict with the local level, some interviewees (10, 18 and 31) said that the national level did offer local policymakers sufficient funding for sustainable mobility. For instance, Nijmegen received funds from the national government to build two new train stations (in 2013 and 2014), thus stimulating travellers to choose the train (interviewee 31). Another interviewee from the national level said that

> *When they [cities] do the things in line with our national mobility policy, they will get the money that's part of those policies. And when they don't align with it, they don't of course (interviewee 18, national level).*

So, opinions did seem to differ somewhat concerning the main conflict.

This concludes all of the conflicts and synergies that were mentioned by at least three interviewees. We now turn to discuss what these results mean in terms of the multilevel challenges of implementing mobility transitions across Europe, in the Netherlands in particular.

5. Discussion

An important question for urban mobility in Europe is how conflicts can be mitigated and synergies promoted in the multilevel mix of policies concerning sustainable urban mobility transitions. The data presented here are primarily from the Dutch context, and the findings therefore do not necessarily represent situations elsewhere. With this caveat, the following first explores potential ways to anticipate on the various conflicts and synergies identified above. It then discusses the findings in the broader context of sustainability transition research.

5.1. Transforming the focus of funding from infrastructure to broader mobility

Section 4.3 identified the main policy conflict in the multilevel governance of sustainable mobility transition as follows: cities depend on national funding for significant initiatives towards sustainable mobility, but national funding does not really promote sustainable urban mobility. Indeed, national funding in the Netherlands was provided primarily to solve bottlenecks in the current dominant mode of travel, which is car mobility. Thus, national funds went mainly to road infrastructure adaptations and extensions. Mobility policymakers, moreover, complained that funding structures for mobility still expected or prematurely dictated policies in the context of constructing road infrastructures. This raises the question of how this conflict might be mitigated.

In 2011, the Netherlands initiated a programme aimed at better utilizing existing infrastructure. That programme set in motion a gradual (although slow) change in the funding structure. By 2018, publications noted the need to broaden the focus of funding, and avoid it predetermining investments in infrastructure (IandW, 2018). This suggests that expanding the focus of funding from the construction of infrastructures only, to solve bottlenecks, towards achieving a broader set of mobility objectives can bring about more opportunities for recipients to design measures promoting behavioural change (i.a. a modal shift) at the local level. Furthermore, this alternative, 'broader' focus can serve to fund 'innovation projects', such as Urban Living Lab 'experiments', that in turn can produce lessons for how to tailor mobility solutions to local circumstances (Wimbadi et al., 2021; Roebke et al., 2022).[9]

In all countries that struggle with the conflict of funding being biased towards 'solving bottlenecks through infrastructure', such a shift to a broader mobility perspective would seem worthwhile. Where necessary, a pilot programme could be initiated to determine how it might function (e.g., *Beter Benutten* in the Netherlands). The EU is currently exploring this suggestion by making the development of a SUMP a requirement for receiving TEN-T funding. This could be effective, but clearly developments must be carefully monitored to prevent improper or symbolic SUMP creation in order to 'tick the box'. Alternatively, TEN-T funding could incorporate the requirement that projects not only improve car mobility, but also improve alternatives to cars at least to an equal extent.

[9] In the Netherlands this also happened with a number of Mobility-as-a-Service experiments being funded by the MIRT in 2020.

5.2. Institutionalizing multilevel cooperation

Not only are national funding structures poorly aligned with sustainable urban mobility strategies, but the national level, in addition, has significant influence on the local level, while itself being hardly influenceable by the other levels. This is a source of frustration at both the EU and the local level, as both have stronger sustainable mobility ambitions than the national level. How might this conflict be mitigated? The European Green Deal (2019), which sets the main policy direction for the EU's future development, seems to point to the need to give the EU more influence on the member states, including with regard to sustainable urban mobility transition. A limitation, however, is that the societal transformation that the Green Deal pursues is approached primarily in a financial-economic instrumental way, with relatively little sensitivity to national, regional and local circumstances (see Foulds et al. 2023, for a similar critique). This may prevent the EU from overcoming the strong national inertia that has so frustrated many at the local level in the field of urban mobility.

To improve multilevel coordination between the national and other levels, the Dutch national-local collaborative programme on urban area development ('MoVe', mentioned above) may provide an example. This would entail that, for instance, for large urban development projects, a team of local civil servants would need to cooperate closely with representatives from regional, national and European level in order to define a project that both promote sustainable urban mobility and fit the broader policy framework. Furthermore, the EU's Urban Agenda (2016) sets the stage for a new multilevel working method promoting cooperation between cities, member states, the European Commission and other stakeholders in order to strengthen the urban dimension in EU policy. Although these types of cooperation may make the development of a project or strategy more complex because they require a common political will among the different levels of collaboration, they can still serve to make the concerns of each of the participants understood by the others and provide each with a comprehensive understanding of the context. This could act as a magnifier, enabling the levels to see subtle conflicts which are more difficult to detect at a distance and to create synergies.[10]

5.3. Translating climate targets at all levels

Despite the conflicts and misalignments noted above, our findings of consistencies (4.1 section) and synergies (4.3 section) suggest that climate change pressures are raising CO_2 reduction to the highest priority level, next to 'congestion reduction', at all levels of governance. In this sense, the policy aims of the various governance levels are becoming better aligned. While prior to the Paris Agreement, congestion reduction (associated with the economic functioning of cities) was generally prioritized over environmental and liveability concerns, respondents in this research agreed that CO_2 is now seen as equally important as accessibility. For member states in which this is not yet the case, it may be helpful to require that all levels of governance explicitly translate the Paris Agreement for their context and policies. Further elaboration is presented in Colombo and Dijk (2023).

5.4. Contribution to the literature

Our analysis of urban mobility transition, using a conceptual frame connecting synergies and conflicts in multilevel mixes of policieswith shifts towards sustainable mobility modalities, contributes to the literature in two ways. First, whereas the existing transport policy literature focuses on less complex policy designs (i.e., concerning just one modality and one governance level), our study offers insight into the vertical relations of policy mixes for sustainable urban mobility in Europe and discusses potential ways to overcome emerging conflicts. Complementing the few conceptual or empirical studies that analyse policy mixes at the urban level (Curtis and Low, 2012; Givoni, 2014; Buehler et al., 2017; Dijk et al., 2018), our study brings in much-needed insights in the vertical dimension. While urban level studies have highlighted the role of local politics, the obduracy of infrastructure and departmental silo structures within local governments, our analysis shows that urban practices and practitioners can be constrained, as well as, in some cases, enabled, by policy goals, instruments and processes at higher levels of governance.

By doing so, our study offers a second contribution. It pioneers a novel connection between two debates in the sustainability transitions literature: that on policy mixes on the one hand and that on urban transitions, experimentation and learning on the other. This can spark new questions such as how can urban experiments be better aligned to (multi-level) policy mixes to enable sustainability urban transformation? How can urban experiments more effectively lead to policy learning? How may established policies mixes constrain the effect of urban experiments? Ehnert et al. (2018) considered urban experiments (or initiatives, as they called them) in a multilevel governance context, but did not address policy mixes. Zepa and Hoffmann (2023) studied how policy mixes across vertical scales of governance shaped sustainable energy in Latvia, but did not address urban sustainability transitions. We hope that our study triggers more work in this direction, as earlier studies have commented on its relevance for urban transition and called for further research (Hodson and Marvin, 2010; Hodson et al., 2017). The literature on transition policy mixes in general has focused mostly on national and supranational energy issues, using a more extensive conceptual framework that distinguishes coherence from consistency (the former being used with reference to policy processes and the latter with regard to strategy, instrument mix and relations between the two) (Rogge and Reichardt, 2016). Our study is more empirics and practice driven, suggesting that a multi-level policy perspective is required to both understand the practical policy constraints to urban mobility transition to its fullest, and to find effective solutions thereto. Our results identify synergies and conflicts concerning policy goals, instruments and processes, including level of collaboration

[10] At the same time, questions remain over the long-term success of this EU Urban Agenda. It is unclear what happened with the theme (or 'partnership') 'urban mobility' after the final report with recommendations was published by September 2020.

and power aspects. For our purpose of gaining insight into the multilevel, practical challenges of implementing mobility transitions, we believe that our more general definition of coherence was instrumental to enable an integrative approach, as various aspects of coherence emerged in the empirics.

6. Conclusion

Sustainable urban mobility is one of the main challenges facing cities in the EU and requires a complex mix of policies across all the involved levels of governance. To date, however, most existing literature on transport policy tools and goals has focused on the effects of less complex designs, specifically, concerning just one instrument, modality or governance level. The few studies that do analyse or offer concepts for analysis of the more complex transport policy mixes have primarily addressed the urban level (Dijk et al., 2018; Legacy et al., 2012; Givoni et al., 2013; Givoni, 2014; Buehler et al., 2017). Therefore, the effect of policies across multiple governance levels has remained underexposed. This research attempted to fill this gap, offering clearer insight into the main conflicts and synergies between policies at the various layers of governance concerning sustainable urban mobility and exploring how these may be overcome.

We found that climate change ambitions are bringing about greater alignment concerning sustainable urban mobility policies across the various governance layers and that EU funding is well aligned with and supportive of local efforts. At the same time, two key multilevel governance conflicts for sustainable urban mobility transitions were identified: (1) National funding has remained biased towards solving bottlenecks through infrastructure, which however, does not support local level attempts towards sustainable mobility. (2) The national level has significant influence on the local level, while itself being hardly influenceable by the other levels. The latter applies especially to the local and regional level, and to some extent also to the European level. As ways to overcome these conflicts, this research proposed transforming the focus of funding from infrastructure to broader mobility and institutionalizing multilevel cooperation. Implementing these could improve policy outputs and outcomes, enabling more progress to be made towards sustainable urban mobility. This objective is of vital concern to European citizens. It is crucial, therefore, that policies to achieve it are promptly further explored, designed and implemented. For our first recommendation, national governments are the most logical actor to take the lead, while for our second we believe that local governments are in the best position to engage in multilevel cooperation when initiating larger urban development projects.

Our study only scratched the surface of the connection between (multilevel) policy mixes and urban transitions. There is thus ample room for further work. First, as noted, a limitation of this study is its focus mainly on the context of the Netherlands. Each EU member state has a unique governance context and associated policy conflicts. Studying other countries would broaden and deepen the insights gathered here and allow for more tailored recommendations for other countries.

Second, the current analysis included only conflicts and synergies that were mentioned by three or more respondents. This means that we left out a variety of observations, some of which may have been equally valuable. Moreover, at times opinions did seem to differ somewhat concerning the identified conflicts. We sought to maximize transparency regarding our interpretations by providing many quotes and including one or more sentences rather than selecting just a few words. Although we do think our method did lead to highlighting the most important conflicts and synergies mentioned across the 32 interviewees, we acknowledge that a level of interpretation on our part has remained. We also acknowledge that a larger number of respondents would have made our approach to scoring the degree of significance of a reported conflict or synergy more robust (i.e., we estimated the degree of significance by the number of respondents who mentioned it). However, we believe the results still carry weight in indicating the relative importance of the conflicts and synergies and provide a useful basis for future research to build on.

Lastly, our analysis of vertical relations would benefit from being combined with analysis of horizontal relations, meaning conflicts and synergies *within* a particular level of governance. Indeed, it is ultimately the combination of the two that will shape the reconfiguration of urban mobility.

CRediT authorship contribution statement

Xu Liu: Data curation, Formal analysis, Methodology, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review & editing, Investigation. **Marc Dijk:** Conceptualization, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing, Investigation. **Carlo Colombo:** Funding acquisition, Project administration, Resources, Validation, Writing – review & editing, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

Funded by the Maastricht Working on Europe and the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Commission. Neither the European Union nor the European Commission can be held responsible for them.

Appendix A. The list of interviewees

Level	#	Organization	Work Position and Contents	Duration (minutes)	Date
EU (10)	1	Local Governments for Sustainability (ICLEI)	Coordinator Sustainable Mobility and Transport, the coordinator of SUMP	98	26/06/2020
	2	European Cyclists' Federation	Co-CEO of the European Cyclists Federation	46	01/07/2020
	3	Directorate-General for Environment (DG ENV)	Working as a second national expert, sent by the German Ministry of Transport to gain some working experience in the commission	64	28/03/2021
	4	Eurocities	Senior project coordinator, coordinating the EUROCITIES Mobility Team, proposal development, project management, corporate representation and strategy	93	07/07/2020
	5	The Directorate-General for Mobility and Transport (DG MOVE)	Deputy director general in DG MOVE, European Coordinator for Road Safety and Sustainable Mobility	41	08/07/2020
	6	DG MOVE	Dealing with innovation research of urban mobility	71	05/03/2021
	7	The Directorate-General for Regional and Urban Policy (DG REGIO)	Team leader for all transport mobility related issues	94	05/02/2021
	8	DG ENV	Urban environment Policymaking, coordination with different departments on urban environment policy	86	06/01/2021
	9	Cities and Regions for Transport Innovation (POLIS)	Director of POLIS (Cities and Regions for Transport Innovation), initiating research activities, look at policy agenda for cities and the governance issues	70	07/07/2020
	10	European Institute of Innovation and Technology (EIT) Urban Mobility	EU affairs manager, funding horizon and innovation researches to help cities conceptualize a sustainable mobility plan	71	03/12/2020
National (9)	11	Friends of the Earth Netherlands (Milieudefensie)	Campaigner on traffic and climate justice.	57	05/01/2021
	12	Nature & Environment (Natuur & Milleu)	Program leader of mobility and market	89	23/02/2021
	13	Urban Mobility at German Institute of Urban Affairs, former employee of POLIS	Team leader of urban mobility at German Institute of Urban Affairs, heads the "Local Mobility" team in the mobility research and the bicycle academy, which connects large and small municipalities from all over Germany and makes them fit for more and better cycling.	49	26/11/2020
	14	Netherlands Environment Assessment Agency (Planbureau voor de Leefomgeving)	Urban Planner	77	07/11/2020
	15	Ministry of Infrastructure and Water Management	Director of Mobiliteit en Gebieden Program	51	16/08/2021
	16	Ministry of Infrastructure and Water Management	Expert in developing mobility plan, link between the expert and the policymaker, the minister and our management.	69	23/09/2020
	17a&b	Ministry of Internal Affairs	a. Coordinator of spatial planning for the South Holland province b. Building bridges between knowledge and policymaking, making new connection, high quality public transport	85	17/12/2020
	18	Ministry of Internal Affairs	leading policy integration and science in the National environmental vision development, coordination of making the report	71	12/11/2020
Regional (7)	19	Rotterdam–The Hague metropolitan area	Project manager for Mobiliteit bij Metropoolregio Rotterdam–The Hague	76	05/01/2021
	20	Province Zuid Holland	Urban geographer, urban development, regional developments	66	16/12/2020
	21	Province Limburg	Senior policy advisor on smart/sustainable mobility, arranging investments and finding the corporations and some mutual interests with the other parties from different levels of governance	68	01/12/2020
	22	Province Gelderland	Regional mobility coordinator	86	29/01/2021
	23	Arnhem Nijmegen Region	Sustainable urban mobility development, connect with EU	92	08/10/2020
	24	Arnhem Nijmegen Region	Program manager on transport, sustainable mobility, accessibility in the Arnhem Nijmegen city region	84	12/02/2021
Local (6)	25	Maastricht Bereikbaar	Policy researcher and advisor, Cluster Mobility & Traffic, Programming regional mobility policy (North and Central	72	22/09/2020

(*continued on next page*)

(*continued*)

Level	#	Organization	Work Position and Contents	Duration (minutes)	Date
			Limburg & West Brabant), Monitoring & Evaluation Roermond Accessibility, Research into the introduction of e-scooter sharing for GoedopWeg & Municipality of Utrecht		
	26	Rotterdam City	Program manager, organizing programs of the mobility department	85	03/12/2020
	27	Department of Sustainability, Rotterdam City	Work together with the department of mobility for sustainable urban mobility development, responsible for the climate agreement, work on the zero-emission mobility policy	82	17/12/2020
	28	Rotterdam City	Coordination, policy advisor, more focus on freight transport and city logistics	80	30/09/2020
	29	Maastricht City	Mobility policymaker, urban planner	89	18/09/2020
	30	Maastricht City	Project leader of Omgevingsvisie (spatial planning and urban development vision)	74	07/08/2020
	31	Nijmegen City	Program Manager of Smart and Clean on The Road	95	12/06/2020

Appendix B. Interview format

1. How do you view or define the concept of sustainable urban mobility at your department or organization?
2. What is your role at your department or organization concerning urban mobility?
3. What do you see as the main policies or instruments at your governance level affecting sustainable urban mobility?
4. What do you see as the main policies at other governance levels (EU, national, regional and local) in shaping sustainable (urban) mobility?
5. How do instruments interfere with each other? For example, conflicting or helping each other.
6. Who do the instruments primarily affect: travellers (i.e. their capabilities or perspectives), local planners (idem), and public transport operators, parking operators or vehicle sharing operators, local infrastructures?
7. How do these policies and incentives play out in cities? Do you see specific differences across cities and/or countries, or do you mainly see recurring patterns? If so, what are these differences and similarities?
8. What are the main (governance) challenges in effectively promoting sustainable (urban) mobility? Who is to blame for the lack of success? Do you see more a problem of inertia by people, conflicting policies or both?
9. In your view, what should be done to promote sustainable urban mobility by each level? Who should take the lead in this?
10. In your view, who else should we talk to?

Appendix C. The references and sources of the related policy instruments

Policy instruments	Sources (all the links are acquired on 16th February 2023)
EU Green Deal	EC 2019. The European Green Deal. https://ec.europa.eu/commission/presscorner/detail/en/ip_19_6691
Sustainable Urban Mobility Plan	EC 2013. Sustainable Urban Mobility Plan. https://www.eltis.org/guidelines/second-edition-sump-guidelines
EU Air Quality Directive	EC 2022. EU Ambient Air Quality Directives. https://environment.ec.europa.eu/topics/air_en
CLean and Better Transport in Cities (CIVITAS)	EU Horizon 2020. CLean and Better Transport in Cities. https://civitas.eu/projects
EU White Paper on Transport	EC 2011. White paper 2011. https://transport.ec.europa.eu/white-paper-2011
Urban Mobility Package	EU 2013. Together towards competitive and resource-efficient urban mobility. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52013DC0913
Urban Agenda	EC 2016. Urban Agenda. https://futurium.ec.europa.eu/en/urban-agenda/pages/what-urban-agenda
Smart and Sustainable Transport Strategy	EU 2020. Sustainable and Smart Mobility Strategy. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52020DC0789
Regional Development Funds	EC 2021. European Regional Development Fund. https://ec.europa.eu/regional_policy/funding/erdf_en
EU Car Emission Regulation	EC 2019. CO2 emission performance standards for new passenger cars and for new light commercial vehicles. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A02019R0631–20210301
EU Emission Norms	EC 2016. Euro 6. https://eur-lex.europa.eu/eli/reg/2016/646/oj
EU Infrastructure Funding (TEN-T)	EC 2013. Trans-European Transport Network. https://transport.ec.europa.eu/transport-themes/infrastructure-and-investment/trans-european-transport-network-ten-t_en
EU Alternative Fuel Infrastructure	EC 2014. Directive 2014/94/EU of the European Parliament and of the Council on the deployment of alternative fuels infrastructure. https://www.fao.org/faolex/results/details/en/c/LEX-FAOC161135/
Public Service Obligations	EC 2007. Public service obligations. https://transport.ec.europa.eu/transport-themes/public-service-obligations_en
EU 100 Intelligent City Challenge	EC 2020. 100 Intelligent City Challenge. https://www.intelligentcitieschallenge.eu/

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Policy instruments	Sources (all the links are acquired on 16th February 2023)
EU Driving Urban Transition Partnership	EU Horizon 2020. European Partnerships. https://www.era-learn.eu/partnerships-in-a-nutshell/european-partnerships/general-information
Green City Award	EC 2008. European Green Capital Award. https://environment.ec.europa.eu/topics/urban-environment/european-green-capital-award_en
Mobility Week Campaign	EC 2002 till now. European Mobility Week. https://mobilityweek.eu/the-campaign/
Open Data Directive	EU 2019. Directive 2019/1024 of the European Parliament and of the Council on open data and the re-use of public sector information. https://eur-lex.europa.eu/legal-content/EN/TXT/?qid=1561563110433&uri=CELEX:32019L1024
Klimaatakkoord	NCA 2019. National Climate Agreement. https://www.klimaatakkoord.nl/documenten/publicaties/2019/06/28/national-climate-agreement-the-netherlands
Omgevingswet Agenda	GONL 2023. Omgevingswet. https://www.rijksoverheid.nl/onderwerpen/omgevingswet
Program Mobility as A Service	I&W 2017. Mobility as a Service. https://www.rijksoverheid.nl/onderwerpen/mobiliteit-nu-en-in-de-toekomst/mobility-as-a-service-maas
Multiyear Fund Infrastructure, Space, and Transport Program (MIRT)	I&W 2018. The Dutch Multi-Year Programme for Infrastructure, Spatial Planning and Transport (MIRT). https://www.government.nl/documents/leaflets/2018/02/07/the-dutch-multi-year-programme-for-infrastructure-spatial-planning-and-transport-mirt-summary
NOVI (national spatial vision)	BZK 2020. Nationale Omgevingsvisie. https://denationaleomgevingsvisie.nl/default.aspx
Program MoVe	Leerplatform MIRT. https://leerplatformmirt.nl/themas/adaptief-programmeren/move-programma/
Smart Mobility (program)	I&W 2016. Smart Mobility. https://www.rijkswaterstaat.nl/en/mobility/smart-mobility
Logistical Laws	Transport & Logistiek. https://transportlogistiek.nl/branche/wet-regelgeving/
Car Taxation	Belastingdienst 2023. https://www.belastingdienst.nl/wps/wcm/connect/bldcontentnl/belastingdienst/prive/auto_en_vervoer/belastingen_op_auto_en_motor/motorrijtuigenbelasting/
Charging Infrastructure	RVO 2020. The National Charging Infrastructure Agenda. https://english.rvo.nl/sites/default/files/2020/10/Factsheet%20The%20National%20Charging%20Infrastructure%20Agenda.pdf
National Zero Emission Bus	I&W 2016. National Administrative Agreement Zero Emission Buses (BAZEB). https://www.interregeurope.eu/good-practices/national-administrative-agreement-zero-emission-buses-bazeb-in-2016-in-the-netherlands
Beter Benutten Program	I&W 2012–2018. Beter Benutten. https://www.rijksoverheid.nl/documenten/brochures/2013/09/20/beter-benutten
POVI (provincial spatial vision)	Provincie Limburg 2021. De Omgevingsvisie Limburg. https://www.limburg.nl/onderwerpen/omgeving/omgevingsvisie/
Parking Norms	I&W 2019. Parking Policies in the Netherlands. https://ecf.com/sites/ecf.com/files/ECF%20-%20Parking%20in%20the%20Netherlands_0.pdf
Car Sharing	I&W 2021. Deelauto- en deelfietsmobiliteit in Nederland file:///C:/Users/m.dijk/Downloads/Deelauto-+en+deelfietsmobiliteit+in+Nederland-Ontwikkelingen,+effecten+en+potentie-pdfA.pdf
Concession for Electric Vehicles	I&W 2023. Subsidieregeling elektrische personenauto's particulieren. https://wetten.overheid.nl/BWBR0043600/2023–01–01
Tendering process public transport operator	Concessions and tenders. https://www.government.nl/topics/mobility-public-transport-and-road-safety/public-transport/consessions-and-tenders
South Limburg Bereikbaar	Funded by the national Beter Benutten programme. South Limburg Bereikbaar. https://www.zuidlimburgbereikbaar.nl/en/about-us/
Fast Bike Lane	Rijkswaterstaat 2023. https://rwsduurzamemobiliteit.nl/@209367/factsheet-fietsinfrastructuur/
City Climate Agreement	Rotterdam municipality 2019. Rotterdam Climate Agreement. https://cdn.locomotive.works/sites/5ab410c8a2f42204838f797e/content_entry5ab410faa2f42204838f7990/5be174d6337f770010c1b69f/files/1.2.2_Rotterdam_Climate_Agreement_ENG.pd
Sustainable Procurement Policy	RIVM 2015. Sustainable procurement. https://www.rivm.nl/en/topics/sustainable-procurement
GOVI (Municipal spatial vision)	VNG 2023. https://vng.nl/artikelen/gemeentelijke-omgevingsvisie
Stedelijk Verkeersplan (Urban Traffic Plan)	Rotterdam municipality 2017. Rotterdam Stedelijk Verkeersplan. https://mkb-rotterdam.nl/wp-content/uploads/2021/09/Stedelijk-Verkeersplan-Rotterdam-20170123_compressed.pdf
Zero Emission Logistics Agreement	I&W 2021. New agreements on urban deliveries without CO2 emission. https://www.government.nl/latest/news/2021/02/11/new-agreements-on-urban-deliveries-without-co2-emission
Low Emission Zone	Amsterdam municipality 2008. Low Emission Zone. https://www.rai.nl/en/route/low-emission-zone/
P&R (Park and Ride)	Maastricht Bereikbaar. $P + R$ facilities. https://www.maastrichtbereikbaar.nl/en/info-for-travellers/car/pr-facilities-in-limburg/
E-hubs	Nijmegen Municipality. eHubs. https://www.nijmegen.nl/nieuws/nieuwsdossiers/dossier-nijmegen-goed-op-weg/deelvervoer/
Intelligent Traffic Lights	I&W. Intelligent Traffic Light Installations. https://www.rijkswaterstaat.nl/en/mobility/smart-mobility/talking-traffic/intelligent-traffic-light-installations

Appendix D. Descriptions of main mentioned policies

Main EU Policies	The European Green Deal is *'a roadmap for making the EU's economy sustainable by turning climate and environmental challenges into opportunities across all policy areas'* (EU, 2019), which plays the role in deriving projects and policy measures to promote sustainability (interviewee 1, 3, 8). One interviewee from DG MOVE argues that the European Green Deal is a *'bigger game changer than the Paris Agreement as the Paris Agreement was not legally binding'* (interviewee 6). Though the Green Deal leads the member states do a lot of work in terms of reaching the CO_2 reduction goal, it has not trickled some of the ministries that kept doing what they always do, such as prioritizing on economic growth and subsidizing gas vehicles (interviewee 14).

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	SUMPs (Sustainable Urban Mobility Plan) issued by the European commission have been designed to solve mobility-related problems in city regions more efficiently. It guides and encourages cities to develop sustainable urban mobility plans, share best practices, and showcase a series of successful projects (interviewee 5). Most of the interviewees who mentioned it and rated it as a successful policy in integrating sustainability into the long-term mobility plans are from EU (interviewee 4, 5, 6, 7, 8, 10). The key reason pointed by the project coordinator of Eurocities is: *'it sets the example for other smaller cities around these cities that made these plans, like Amsterdam…If they make these steps to more sustainable development of the city, then it helps set the example for the other cities. That helps it to go top-down from Europe'* (interviewee 4). Whereas the mobility expert from South-Limburg alluded that the SUMPs did not stimulate them to do much in corresponding with the SUMT (interviewee 25). Moreover, a small group of cities, especially the small size cities, were not satisfied with this measure since they cannot get the specific information for their type of context. But the EU does not have enough capacity to tackle such complaints, specified by the coordinator of SUMPs (interviewee 1). DG ENV introduced a European Commission directive aimed at improving local air quality in 2008, the EU Air Quality Directive. Although EU norms for particulate matter had gone into force in 2005 already, they had resulted in little action in EU member states (interviewee 5, 8). Since the issue of the Air Quality Directive, the Netherlands acted immediately with a national plan to improve air quality in 2009, including the promotion of e-mobility (Ministry of Infrastructure and the Environment, 2009). Both the interviewees from DG REGIO and DG ENV rated the Air Quality Directive as the most powerful EU instrument in promoting SUMT: *'I think the most powerful driver has been the air quality legislation. Because I mean, if people get taken to court, because they are, again, not meeting the thresholds. I think this has been a strong driver' (interviewee 7, E).*[1] '*I would say the air quality legislation in relation to, is probably one of the most the biggest impacts on urban mobility that we have. Because the SUMP is not compulsory. I think cities are supposed to have they're maybe supposed to, if they don't, they're not going to be taken to court over it' (interviewee 8, E).* Although the Air Quality Directive is a strong incentive for SUMT, some interviewees (3, 8, 11, 23) pointed that cities are still far from the air quality standard set by WHO (World Health Organization), so the EU standard definitely needs to become stricter. Trans-European Transport Network (TEN-T) creates an integrated and intermodal long-distance, high-speed, trans-European transport network, mechanized by providing funding for infrastructure construction including all modalities (e.g. road, rail, water, air). TEN-T funding is granted under a system of centralized administration. That means that the Commission sets the rules. Sustainability is, currently not a criterion of importance in funding decisions. Infrastructure is developed based on the need for more capacity (usually due to congestion issues), not to support modal shift to more sustainable modalities (interviewee 5, 10). For instance, when funding new ring roads or road tunnels, induced demand of car mobility is not considered. The lack of sustainability consideration is also noted in a recent evaluation on the TEN-T framework (interviewee 3). Increasingly, however, there are calls to include sustainability requirements (such as the existence of a sustainable mobility plan), in order to receive TEN-T funding (interviewee 28, 31).
Main national policies	The Netherlands National Climate Agreement (NCA, 2019), involving over 100 parties (governments, the business community, social organizations), aims to reduce the greenhouse gas emissions by 49 % in 2030 and 95 % in 2050. For the mobility sector, the key measures include the shift to zero-emission cars, the introduction of low-emission zones in cities, a form of road pricing, and more attention for alternatives to the car. A key financial instrument of the Dutch national government is the Multi-Year Programme for Infrastructure, Spatial Planning, and Transport (MIRT), managed by the Ministry of Infrastructure and Water Management. This program earmarks large sums of money to address 'bottlenecks' (i.e. apparent infra capacity issues that lead to congestion) for the coming decades. In some way it is the national counterpart of the TEN-T. The MIRT is important for local and regional levels of governance because large investments are impossible without funding from MIRT (interviewee 17, 20). However, the emphasis on infrastructure also tends to bias policy strategies towards 'expanding infrastructure' (which is prone to lead to induced demand), as opposed to more demand management policies (interviewee 14). Local authorities (generally in need of money from supra-local level) are incentivized to reframe their policies into infra-driven policies. In the last five years, possibly under influence of climate ambitions, there are efforts to gradually reframe the fund from an 'infrastructure' into a 'mobility' fund (interviewee 12, 31). Taxation plays a strong role in speeding sustainable mobility transitions. Like most countries, Dutch car drivers were paying various types of taxes: a monthly road tax, fuel tax for every liter bought, VAT (Value Added Tax) when the vehicle was purchased, and, for lease drivers, an addition to their income tax ('bijtelling' in Dutch). These taxes on car mobility were slightly higher than average in Europe. For traditional fossil-fuel vehicles, VAT had been connected to energy labels since 2006, but this became combined with CO2 emissions in 2010, and fully based on CO_2 after 2013. After 2010, a 'feebate' system gave low-emission vehicles a tax discount (or even exemption), while high-emission vehicles were taxed extra, so they were taxed progressively according to their CO2 emissions. In addition, there was a generous tax exemption for full battery-electric vehicles. In the policy communities, one spoke of 'greening of the tax regime'. Still, since tax is not based on km's driven, there was little incentive to constrain total distance driving (interviewee 15).
Main regional policies	Dutch provincial governments have the authority to provide the concessions for public transport operators (bus, regional trains, trams, metro). The regulation of the regional public transport not only includes adjusting the amount of buses or the energy transitions for the busses, but also contains drawing new lines and amending the frequencies based on travellers' needs. For instance, the possibilities of more direct, more frequent and faster public transport, to make sure that there is always an alternative for the car use, which on the other hand smooths the mobility transitions (interviewee 22).
Main local policies	Local governments have the authority over the public space (i.e. incl. all road space) within the municipal boundaries, in line with the subsidiarity principle. Accordingly, local traffic and land use policies shape the mobility infrastructure, and have a strong influence on the space for car roads, bicycle paths, bus lanes, pedestrian spaces, parking spaces and tariffs. Clearly these decisions strongly shape the attractiveness of the various mobility practices (i.e. car mobility, cycling, public transportation, walking). Zero or Low Emission Zones have been introduced in 2010s by some Dutch cities based on local needs, which lead to a patchwork of different rules. The national government intervened and since 2019 with a national standard was set. Cities were not always happy with this (interviewee 11, 23, 26, 29, 30). Maastricht, for instance, close to the German border, had adopted a system adapted to Germans guests, but was now required to change to much more expense national one, which did not receive sufficient support in the council (interview 29). Concerning freight traffic, Dutch cities had been cautious for strict measures, fearing commercial competition from other cities. The participatory process around the Dutch climate agreement, however, resulted in an agreement between the largest 40 cities to achieve a zero-emission zone for freight by 2025, which is listed as a local policy instrument and has been mentioned the most in the local level.

[1]In the quotes in this table we use E for the abbreviation of European level, N for national level, R for regional level, and L for local level.

Appendix E. All reported contradictions and synergies in promoting sustainable urban mobility development

Contradictions (-)	EN	ER	EL	NR	NL	RL	Synergies (+)
Lack of money (7[a])					L^{-}[b]		Provide funding (12)
		E^{++}	E^{++++}	N^{+}	N^{+++}	R^{++}	
Funding mainly for (car/road) infrastructure (6)				N^{--}	N^{--}		Subsidy for SUM projects (4)
			E^{+}	N^{+}	N^{+}	R^{+}	
Funding still for promoting car use (3)			E^{-}		N^{--}		Offer SUM projects (6)
	E^{+}	E^{+}	E^{+}	N^{+}	N^{+}	R^{+}	
Have different targets (6)	E^{--}			N^{-}	N^{--}		Set common goals (15)
	E^{+}	E^{++}	E^{+}	N^{+++}	N^{++++}	R^{++++}	
Lack of influence/power (10)	E^{--}		E^{-}		L^{-}[c]	R^{-} L^{-}	Set guidelines (1)
			E^{+}				
Lack of information (1)			L^{-}				Good networks (2)
			E^{++}				
Lack of knowledge and capacity (2)			L^{-}			L^{-}	Provide trainings and webinars (2)
			E^{++}				
Lack of adaptability (3)		E^{-}	E^{-}			R^{-}	Prompt process for SUMD (2)
	E^{+}	E^{+}					
Lack of communication (5)		E^{-} R^{-}			N^{-}	R^{-} L^{-}	Good cooperation (16)
				N^{+++}	N^{+++}	R^{++} L^{++}	
				R^{+++}	L^{+++}		
Not practical (1)			E^{-}				Evaluating the implemented projects (1)
				R^{+}			
Conflicts with legislation (2)					N^{--}		Ambitious (3)
			L^{++}		N^{+}		

[a]: the number of '-' or '+' were mentioned by different interviewees; [b]: the number of '-' and '+' represent the corresponding contradictions and synergies mentioned by how many different interviewees; E: European level; N: national level; R: regional level; L: local level; EN: relations between European level and national level; ER: relations between European level and regional level; EL: relations between European level and local level; NR: relations between national level and regional level; NL: relations between national level and local level; RL: relations between regional level and local level; c: This is mainly reflected by the local level that the national level has too much power to decide what the policy instruments can be implemented instead of the local governments can decide for themselves.

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